

CHM111 Practice Exam 3 on chapter(s): 5, 6

1.  $-377 \text{ kJ}$ . Since  $\Delta H < 0$ , the reaction is exothermic.
2.  $273 \text{ kJ}$
3.  $E \rightarrow A$  ∴ because this transition corresponds to the largest drop in energy levels, resulting in emission of a photon with the highest frequency (and shortest wavelength).
4.  $7.82 \times 10^{-7} \text{ m}$
5.  $60 \text{ zJ}$  ∴ "A" is the level with the lowest energy, AKA the ground state.
6.  $D \rightarrow C$  ∴ this transition corresponds to the smallest energy difference, resulting in emission of a photon with the longest wavelength and lowest frequency.
7.  $1s: n = 1, l = 0, N_e = 2 \dots 2s: n = 2, l = 0, N_e = 2 \dots 2p: n = 2, l = 1, N_e = 6$
8. Emitted ∴ a photon is emitted when an electron transitions to a lower energy level
9.  $-1.04 \times 10^5 \text{ J}$
10. 3 ∴ the number of lines appearing in the spectrum is equal to the number of transitions. Let  $n$  be the number of excited states; the number of transitions is  $n(n + 1)/2$ .
11.  $A \rightarrow F$  ∴ this transition corresponds to the largest energy difference, resulting in absorption of a photon with the shortest wavelength and highest frequency.
12. (lowest frequency) red light < orange light < yellow light < ultraviolet light (highest frequency)
13.  $1.68 \times 10^{-6} \text{ m}$  ∴  $\Delta E = \left( \frac{-R_y}{n_2^2} \right) - \left( \frac{-R_y}{n_1^2} \right)$  and  $\Delta E = \frac{hc}{\lambda}$
14. 10 ∴ the number of lines appearing in the spectrum is equal to the number of transitions. Let  $n$  be the number of excited states; the number of transitions is  $n(n + 1)/2$ .
15. (lowest energy) red light < orange light < green light < gamma rays (highest energy)
16.  $A \rightarrow B$  ∴ because this transition corresponds to the smallest energy difference, resulting in absorption of a photon with the lowest frequency (and longest wavelength).
17.  $B \rightarrow A$  ∴ because this transition corresponds to the smallest drop in energy levels, resulting in emission of a photon with the lowest frequency (and longest wavelength).
18. (shortest wavelength) ultraviolet light < blue light < orange light < microwaves (longest wavelength)
19.  $6 \text{ J}$
20.  $2.3\%$  ∴  $\Delta x \Delta p \geq \frac{h}{4\pi}$  and  $\Delta p = m\Delta v$
21.  $A \rightarrow B$  ∴ this transition corresponds to the smallest energy difference, resulting in absorption of a photon with the longest wavelength and lowest frequency.
22.  $A \rightarrow G$  ∴ because this transition corresponds to the largest energy difference, resulting in absorption of a photon with the highest frequency (and shortest wavelength).
23.  $3.31 \times 10^{-6} \text{ m}$  ∴ the energy of the emitted photon is equal to the difference in energy levels and  $E = hc/\lambda$
24. Yes ∴ the de Broglie wavelength  $\lambda = \frac{h}{mv}$  is larger than the width of the object.

25.  $140 \text{ zJ}$  ∴ "B" is the level with the second-lowest energy, AKA the first excited state.
26.  $1.65 \times 10^3 \text{ zJ}$  ∴ "C" is the level with the third-lowest energy, AKA the second excited state.
27.  $0.039 \text{ J/g/K}$
28. 1s, 2s, 3s, 3p, 4d
29. 3.385 m
30.  $C \rightarrow A$  ∴ this transition corresponds to the largest energy difference, resulting in emission of a photon with the shortest wavelength and highest frequency.
31. 130 kJ
32.  $200 \text{ zJ}$  ∴ the energy of the absorbed photon is equal to the difference in energy levels.  $|700 - 900| = 200$